National Football League Game Officials Self-Rating of Knowledge in Neuro-Ophthalmic Principles and Practice: A Pilot Program to Improve Precision and Accuracy of Game Official Calls

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Background: To determine whether a neuro-ophthalmic curriculum would improve National Football League (NFL) game officials' self-rated knowledge and interest in neuro-ophthalmic principles to improve precision and accuracy of NFL play-calling.

Methods: The formalized and structured neuro-ophthalmic principles (NOP) curriculum was introduced to 121 NFL game officials, 17 replay officials, and 4 officiating staff who attended the NFL Official Training Camp in Irving, Texas, on September 8 and 9, 2023. Before and after the lecture and videos were introduced, participants completed an optional hard-copy feedback form pertaining to self-reported NOP knowledge, likelihood of using said terms, and interest in future content of NOP applicable NFL officiating. Paired 2-tailed *t* tests were used for statistical analysis to directly compare the self-reported knowledge before and after the neuro-ophthalmic curriculum introduction.

Results: One hundred forty-two participants completed the prelecture and postlecture feedback forms self-reported knowledge after the NOP curriculum was given to the NFL officiating staff. All (142/142) participants completed a survey. There was a statistically significant improvement in the mean ratings of the prelecture vs. postlecture understanding of the specific neuro-ophthalmic terms pertinent to NFL game officials (2.6 [95% CI, 2.3–3.0] vs. 7.9 [95% CI, 7.6–8.2], $P < 0.001$) and 2.7 [95% CI, 2.3–3.0] vs. 7.7 [95% CI, 7.4–8.0]), respectively. There was a statistically significant greater likelihood of using said terms prelecture vs. postlecture (2.9 [95% CI, 2.4–3.4] vs. 7.5 [95% CI, 7.2–7.9], $P < 0.001$).

Conclusions: This study found a statistically significant improvement in neuro-ophthalmic knowledge and a greater likelihood of using NOP terms following the NOP curriculum. NFL game officials, replay officials, and staff are interested in expanding their knowledge in the vision science of neuro-ophthalmic concepts and applications involved in play-calling. We hope that our pilot data will lead to a model of education that will improve the precision and accuracy of NFL play-calls by officials on game days.

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The intersection of neuro-ophthalmology, visual processing principles, and American football game officiating lies at the heart of accurate and precise play assessment. "Making the call" is a challenging process and often a high-stakes, real-time decision, where one call can determine the outcome of a game. The National Football League (NFL) is a multibillion-dollar industry with an average of 16.7 million viewers in 2022. With high financial stakes involved, it is essential to make quick, reliable, accurate play-calls consistently throughout each NFL game. NFL play-calling is a complex task that requires a high visual and cognitive processing demand. NFL officials perform under high scrutiny, high stress, and high-stakes conditions.

Although visual acuity is an important part of accurate play-calling, it is only a small part of the picture. The afferent and efferent pathways in neuro-ophthalmic principles (NOP) play critical roles in visual spatial processing during NFL game play and officiating. Specific neuro-ophthalmic concepts are directly relevant to the responsibilities of NFL game officials including visual acuity, visual

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field, smooth pursuit, saccades, and the vestibulo-ocular reflex.[1]

Currently, training and development of NFL officials inherently and implicitly includes assessment and optimization of afferent input including visual acuity (line of sight, starting and ending position during a play) and visual field as well as efferent input (saccades, pursuit, and vestibular ocular reflex [VOR]) to improve accuracy and precision. Although an extensive NFL game official training curriculum has existed for years in the NFL, Walt Anderson (January 2020), Senior Vice President for NFL Game Officials, hypothesized that developing a formalized vocabulary and curriculum to teach the visual science of game officiating would be helpful to NFL game officials. Although there is a rapidly emerging and expanding literature on visual afferent and efferent processing and performance in professional athletes, little has been written to date regarding professional game officials.[1–7]

A previous pilot study created and introduced a novel NOP curriculum for NFL game officials with prelecture and postlecture surveys completed by 21 of the 26 senior NFL game officials. This pilot study served as a demonstration project for senior NFL game officials and administration, and the preliminary curriculum included information on static visual acuity (SVA), dynamic visual acuity (DVA), minimum position error (MPE), saccadic control, smooth pursuit, VOR, and selective attention deficit. The results of the initial pilot showed that NFL officials had a strong interest in learning more about NOP, and the NOP curriculum improved participants' level of knowledge of NOP.[1]

We describe the larger, follow-up study of the proposed pilot NOP curriculum after implementation for the entire current NFL game official cohort (n = 142). The primary objective was to assess knowledge improvement and interest in the NOP curriculum in a larger cohort to validate and assess the relevance to the target audience. This is the first such study conducted for NFL game officials. We believe that learning the neuro-ophthalmic concepts (i.e., the "why" behind certain principles in the game officials' training curriculum) will enhance game officials' play-calling skills. In addition, a formalized, structured vocabulary may lead to better communication among game officials and an avenue for a standardized method of evaluation of play-calling. These enhanced skills, formalized vocabulary, and systematic method of evaluation may lead to enhanced performance, accuracy, and precision in play-calling.

METHODS

A formal and structured NOP curriculum was initially implemented and tested on senior NFL game officials as a demonstration project before presentation to the entire NFL game official cohort. The lecture materials and video format were then tested on the larger cohort of all 121 NFL game officials, 17 replay officials, and 4 officiating staff (n = 142) who attended the NFL Official Training Camp in Irving, Texas, on September 8 and 9, 2023. Several authors (N.V.C., J.O., and A.G.L.) traveled to Irving, Texas, at the invitation of the NFL to participate in the meeting and to test the NOP curriculum in person on a larger scale. Nonexpert participants (nonmedical investigators) from the National Football League (NFL) front office met with one of the investigators (AGL) and cleared the lecture and questionnaire through regular approval channels at the NFL. During the face-to-face game official meeting in Irvine, Texas, the PI (AGL) explained the purpose of the study, the voluntariness of participation, and the benefits and expected risks (confidentiality protection measures). The curriculum consisted of a one-hour lecture presented in slide presentation format. Short video vignettes containing audio of NOP explanations for specific NFL game officiating calls were also presented. Before the lecture and videos, verbal permission from individuals and group permission from the NFL was obtained to complete the optional hard-copy paper real-time feedback form of 9 questions pertaining to self-reported knowledge on NOP in NFL game officiating.

The data of NFL officials were collected in an anonymous fashion by NFL personnel and the PI staff and not the PI. The NFL has an existing infrastructure and process for evaluation that their officials must participate in. Participants were not required to participate and were given neither incentive nor disincentive for participation. All surveys were both anonymous and confidential with no personal, health, or other identifying information included in the survey. Only aggregated, unidentified, data were collected.

Table 1 summarizes the feedback form containing questions that were provided to the NFL officials to determine improvement in understanding of the NOP terms as pertaining to the NFL game officiating and interest in future content. The scale was scored on a Likert scale of 1 (least) to 10 (most) for each question. If a participant provided a range of numbers for a question, the average of the minimum and maximum numbers divided by 2 was recorded. Anonymous, unidentified data are available on request. The primary outcome measure was NFL game official self-rated knowledge of NOP in relation to NFL play-calling before and after the educational intervention. A paired two-tailed *t* test (2021 Microsoft Excel function application) was used for statistical analysis. We directly compared the self-reported knowledge before and after curriculum introduction. Statistical significance was defined as $P < 0.05$ or better. Analysis was performed for questions 1 and 2, 3 and 4, and 5 and 6, which included both the prelecture and postlecture questions. An average rating of the 10-point Likert scale was obtained for questions 7, 8, and 9.

RESULTS

One hundred forty-two participants completed the prelecture and postlecture feedback forms self-reported knowledge after

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TABLE 1. Prelecture and postlecture feedback form of National Football League game officials and replay staff

1. On a scale of 1 (lowest) to 10 (highest) score, how would you self-rate your level of understanding of ocular terms: "saccade," "pursuit," and "vestibulo-ocular reflex" BEFORE the presentation as they apply to NFL officiating?
2. On a scale of 1 (lowest) to 10 (highest) score, how would you rate your level of understanding of the ocular motor terms "saccade," "pursuit," and "vestibulo-ocular reflex" AFTER the presentation as they apply to NFL game officiating?
3. On a scale of 1 (lowest) to 10 (highest) score, how would you self-rate your level of understanding of terms: "selective attention deficit," "dynamic visual acuity," and "visual field" BEFORE the presentation as they apply to NFL officiating?
4. On a scale of 1 (lowest) to 10 (highest) score, how would you self-rate your level of understanding of terms: "selective attention deficit," "dynamic visual acuity," and "visual field" AFTER the presentation as they apply to NFL officiating?
5. On a scale of 1 (lowest) to 10 (highest) score, how likely will it be that you will use future training in the ocular motor terms such as "saccade," "pursuit," and "vestibulo-ocular reflex" BEFORE the presentation?
6. On a scale of 1 (lowest) to 10 (highest) score, how likely will it be that you will use future training in the ocular motor terms such as "saccade," "pursuit," and "vestibulo-ocular reflex" AFTER the presentation?
7. On a scale of 1 (not relevant) to 10 (highly relevant) score, how would you rate the presentation's relevance to your NFL officiating in real-time games?
8. On a scale of 1 (not relevant) to 10 (highly relevant) score, how would you rate the videos' relevance to your NFL officiating in real-time games?
9. On a scale of 1 (not interested) to 10 (highly interested) score, how interested would be in additional future content on the topics of neuro-ophthalmology as it pertains to NFL game officiating?

NFL, National Football League.

the NOP curriculum was given to the NFL officiating staff. In our survey, 142 of 142 (100%) of participants answered questions 1–7. For one question (#8), 141 (99.3%) of 142 participants answered, and 121 (83.4%) of 142 completed question 9. The NFL game and replay officials' ages ranged from 38 to 69. There were 3 females and 118 males. The prelecture mean for the level of understanding of the NOP terms "saccade," "pursuit," and "vestibular-ocular reflex" in relation to NFL officiating was 2.7 (95% confidence interval [CI], 2.3–3.0) (Fig. 1), and the postlecture of understanding of the same NOP terms "saccade," "pursuit," and "vestibular-ocular reflex" in relation to NFL officiating was 8.0 (95% CI, 7.7–8.3) (Fig. 1).

There was a statistically significant improvement in the mean ratings of the prelecture vs. postlecture understanding of these terms (2.6 vs. 7.9, $P < 0.001$) (Fig. 1). The prelecture mean for the level of understanding of the NOP terms "selective attention deficit," "dynamic visual acuity," and "visual field" in relation to NFL officiating was 2.7 (95% CI, 2.3–3.0) (Fig. 1). The postlecture of understanding of the NOP terms "selective attention deficit," "dynamic visual acuity," and "visual field" in relation to NFL officiating was 7.7 (95% CI, 7.5–8.0) (Fig. 1). There was a statistically significant improvement in the mean ratings of the prelecture vs. postlecture understanding of these terms (2.6 vs. 7.7, $P < 0.001$) (Fig. 1). The preintervention mean score for the likelihood that NFL officials would use ocular motor terms was 2.9 (95% CI, 2.4–3.4). The postlecture mean of the likelihood that NFL officials would use oculomotor terms was 7.5 (95% CI, 7.2–7.9) (Fig. 1). There was a statistically significant greater likelihood of using the designated NOP terms after the intervention compared with before (2.9 vs. 7.5, $P < 0.001$) (Fig. 1).

In this study of the full cohort of NFL game officials in the United States, there was a significant increase in both understanding and more importantly likelihood of using NOP terms after the educational intervention. NFL game officials rated the NOP curriculum as "highly relevant" and "important:" to their officiating in real-time games, with an average rating of 8.8 (95% CI, 8.5–9.1) (Fig. 2). In addition, the videos of plays (NFL Films) demonstrating specific officiating examples of real NFL games with incorporation of NOP terms were found to be highly relevant, with an average rating of 8.8 (95% CI, 8.5–9.0) (Fig. 2). Finally, the NFL officials had an average rating of 8.2 (95% CI, 7.8–8.6) in terms of level of interest in future content on NOP topics pertaining to NFL game officiating.

CONCLUSIONS

Our study is novel and unique and confirms the initial findings of our pilot study in the senior NFL game officials regarding demonstrated interest and audience-rated value in incorporating NOP and practice into the NFL game official training curriculum. We believe that this effort will eventually lead to increased precision and accuracy of NFL play-calling in real time. There was a statistically significant improvement in all 3 self-rated knowledge mean scores of NOP terms after the educational intervention. Moreover, NFL game and replay officials and staff felt the curriculum was highly relevant (8.8 of 10) and interested in expanding their knowledge on these topics (8.8 of 10).

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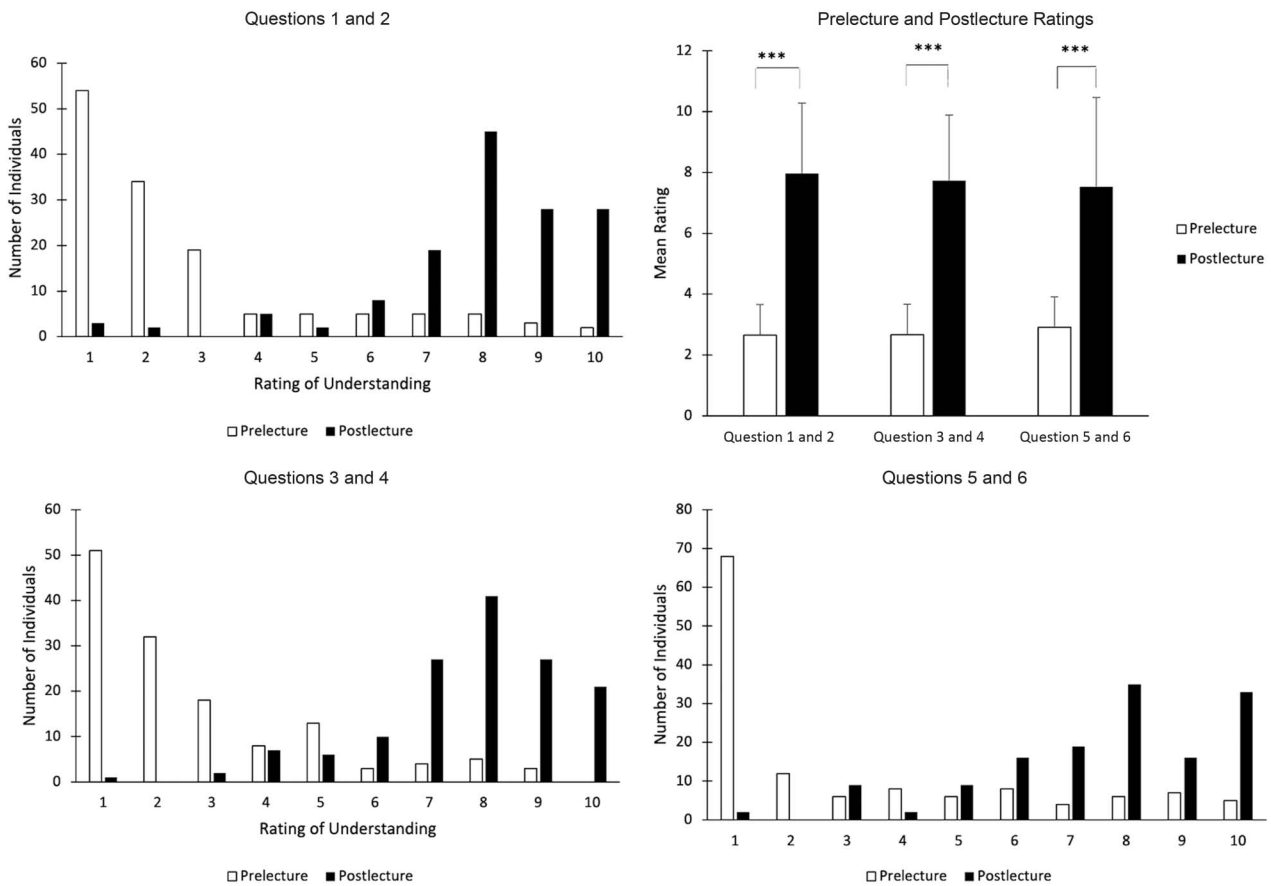

FIG. 1. Graphs depicting survey results for questions 1–6 and paired two-tailed test comparing prelecture vs. postlecture self-rated knowledge of NOP terms. NOP indicates neuro-ophthalmic principles.

We acknowledge that there are limitations to this study. Notably, the total number of NFL game officials available for study is low (n = 122). There were also a few participants with higher baseline (preintervention) scores who had already participated in the pilot study in the previous year. We were unable to identify these individuals or remove them from the data pool because of the methodology of the anonymous and confidential survey instrument. The response rate for question 9 was lower in our study but perhaps was due to the question being on the back of a 2-sided (front and back) paper form (rather than a digital continuous survey format). We chose a traditional paper format, however, to preserve anonymity and to ensure participation without the risk of any personal identifying information.

We hope to continue our work with the NFL and to align traditional and historical NFL game official training with the science of afferent and efferent NOP. We are working with the NFL to analyze play and replay data to determine the potential NOP root causes for an inaccurate play-call, and we have developed a scoring rubric to define specific objective deficiencies based on film in game official line of sight, visual field, saccadic prioritization, pursuit, or dynamic VOR.

These deficiencies translate into formative feedback information to improve initial and postplay development positioning. Current NFL game official instruction already emphasizes the need to reduce unnecessary movement and to focus on statistically likely play evolution (e.g., run vs. pass) in different play situations. Existing game official teaching approaches aim to prevent officials from being out of position or experiencing reduced visual acuity (line of sight) or visual field.

Analysis of hundreds of game situations by the NFL provides important data for defining the best visuomotor strategies for game officials in any given circumstance. For example, in the analysis of offensive fumbles inside and outside of the 30-yard line to the goal, no fumbles occurred within the first-yard line during plays of 30 yards or more and only 15 of total 44,208 plays (0.03%) of plays outside the 30-yard line resulted in a runner being tackled inside the one-yard line (personal communication Walter Anderson, Senior VP NFL). Thus, if there is adequate line of sight (fovea on the target) and adequate visual field, the current recommendation for NFL game officials in this situation is to remain stationary as play develops and avoid running with the player to the goal line. In the past, NFL game officials had been admonished to "not get beat to the goal line." Analysis of the fumble data, however, suggests that this particular football play situation

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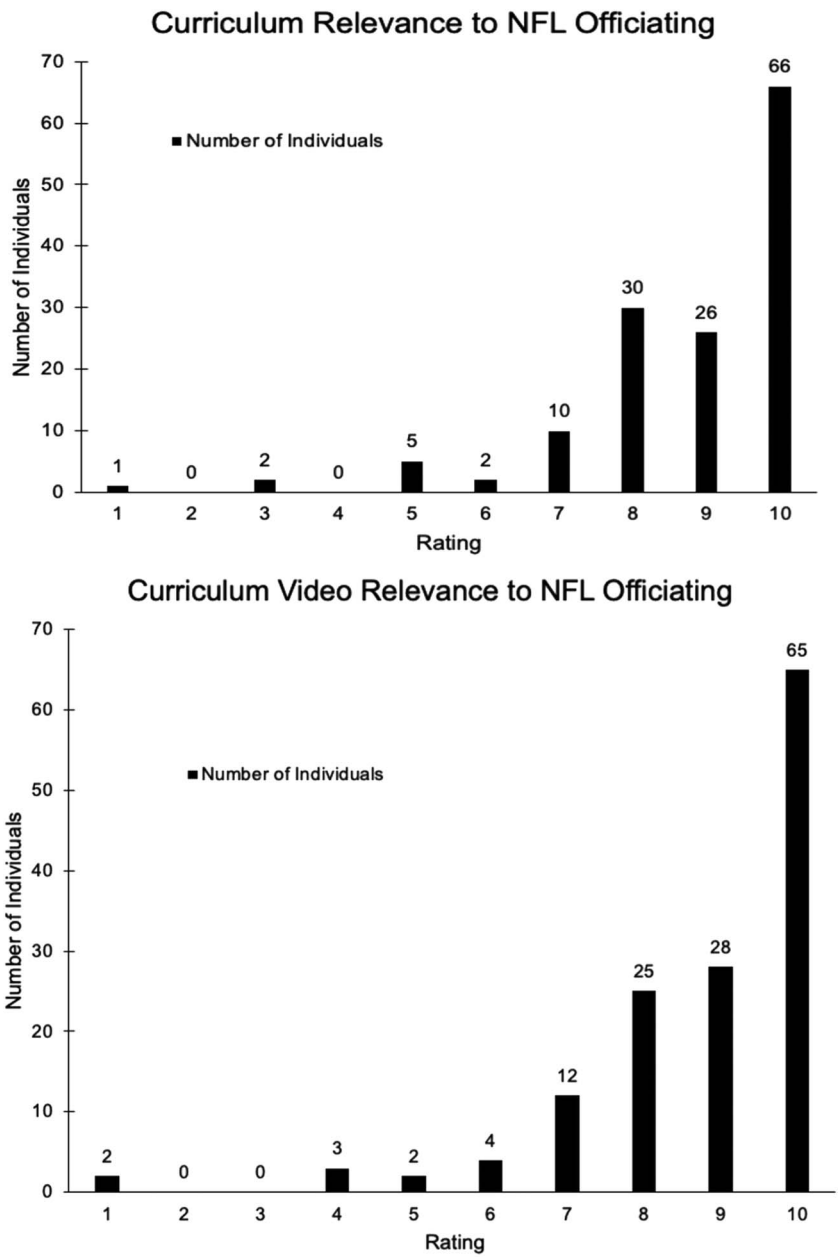

FIG. 2. Ratings of curriculum and video relevance to NFL officiating. NFL indicates National Football League.

does not require that the game official run to the goal line with the player. In fact, dynamic VOR and potential degradation of foveation time might occur by running when a stationary position would be sufficient 99% of the time in fumble situations. By combining NOP and NFL game statistical data, we believe that we can improve the training and development of officiating in the NFL. We also believe that having a shared vocabulary surrounding NOP in the NFL by providing the scientific principles behind the call will enhance play-calling precision and accuracy on game day.

The next phase of our project with the NFL is to develop and test formative and summative knowledge assessment quizzes of NOP using real game videos (NFL Films). Currently, NFL game officials take weekly formative assessments to teach and learn specific game situations in a multiple-choice question (MCQ) format. We are currently piloting a formal scoring rubric and developing MCQ. Future work is necessary to confirm whether these educational initiatives will result in improved precision and accuracy by NFL game officials on game day.

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REFERENCES

1. **Vogt AZ**, Woodland MB, Carter MJ, Lee AG. Curriculum in neuro-ophthalmic principles for national football League game officials: comparison of pretraining and posttraining ratings of knowledge. J Neuroophthalmol. 2023. Online ahead of print.
2. **Ghasemi A**, Momeni M, Jafarzadehpur E, Rezaee M, Taheri H. Visual skills involved in decision making by expert referees. Percept Mot Skills. 2011;112:161–171.
3. **Ziv G**, Lidor R, Zach S, Brams S, Helsen WF. Gaze behavior of referees in sport-A review. Front Sports Act Living. 2020;2:572891.
4. **Aginsky KD**, Noakes TD. Why it is difficult to detect an illegally bowled cricket delivery with either the naked eye or usual two-dimensional video analysis. Br J Sports Med. 2010;44:420–425.
5. **Baptista AMG**, Serra PM, Faisal M, Barrett BT. Association between clinical vision measures and visual perception and soccer referees' on-field performance. Optom Vis Sci. 2021;98:789–801.
6. **Mather GA**. A step to VAR: the vision science of offside calls by video assistant referees. Perception. 2020;49:1371–1374.
7. **de Oliveira MC**, Orbetelli R, de Barros Neto TL. Call accuracy and distance from the play: a study with Brazilian soccer referees. Int J Exerc Sci. 2011;4:30–38.

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